A PROJECTIVE MODULE EMBEDS ITS ELEMENTS IN FINITE FREE SUMMANDS AFTER ADDING A FREE MODULE

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Allowed standard prerequisites (editorial).

- A projective module becomes free after adding a free module. invoked at proof lines 39017-39019. Provides the free module containing F plus P.

Source reference. Original lemma 0GVG: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38996–38996. Cited at proof lines 39017-39018 for the freeness of F + P; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let P be a projective module. Let $s \in P$. There exists a finite free module F and a finite free direct summand $K \subset F \oplus P$ with $(0, s) \in K$.*

Proof. By Lemma 0GVG we can find a (possibly infinite) free module F such that $F \oplus P$ is free. Then of course $(0, s)$ is contained in a finite free direct summand $K \subset F \oplus P$. In turn K is contained in $F' \oplus P$ where $F' \subset F$ is a finite free direct summand. □

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